

HOMEWORK 6 CAPSTONE — GENERATIVE AI FOR SOFTWARE ENGINEERING

Multi-Agent Banking Pipeline

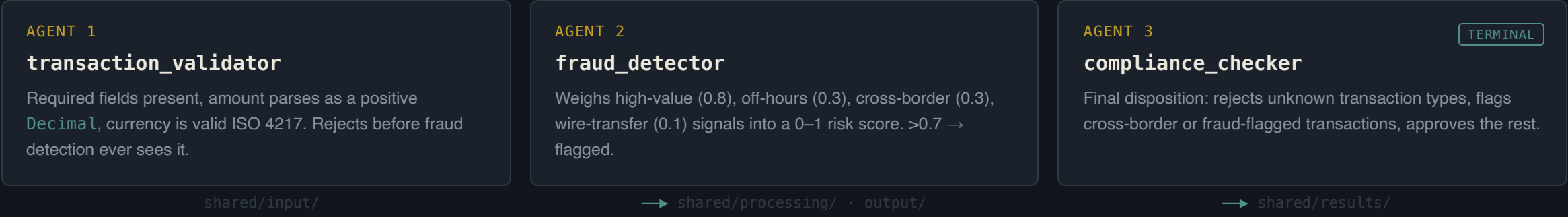
Three cooperating agents validate, risk-score, and clear raw bank transactions through a file-based message protocol — built end to end by four meta-agents: a spec writer, a code generator grounded in live context⁷ lookups, a test generator gated by a coverage hook, and a documentation agent.

Created by Volodymyr Zubchynskyi `sample-transactions.json` → `shared/results/`

validator ● → fraud ● → compliance ●

Three agents, one file-based protocol

Every hop is a JSON message on disk — no direct function calls between agents, no shared process.



integrator.py — orchestrator, drives all three agents in order **message envelope** — `message_id · timestamp · source/target_agent · data`

audit log — `pipeline_run.log`, one JSON line per decision

MCP, skills, and a coverage gate

The pipeline that builds the pipeline: research, control, and enforcement.

MCP SERVERS

`context7`

Live queries against `/prefecthq/fastmcp` and `/python/cpython` while writing `mcp/server.py` — see `research-notes.md`

`pipeline-status`

Custom FastMCP server: tools `get_transaction_status`, `list_pipeline_results` + resource `pipeline://summary`

CLAUDE CODE SKILLS

`/write-spec`

Regenerates `specification.md` from the template

`/run-pipeline`

Clears `shared/`, runs `integrator.py`, summarizes results

`/validate-transactions`

Dry-run validator only, tabulates `accept/reject`

PUSH-BLOCKING HOOK

`PreToolUse` → `Bash`

Intercepts any command containing "git push"

`pytest --cov ≥ 80%`

Runs the full suite first; exit 2 blocks the push below threshold

`coverage-gate.log`

Every allow/block decision is timestamped for audit

51 tests, 99% coverage, 8/8 processed

98.98%

test coverage (gate: 80%)

51

unit + integration tests

8 / 8

sample transactions processed

3 / 3 / 2

approved / flagged / rejected

TRANSACTION	AMOUNT	STATUS	REASON
TXN001	1,500.00 USD	approved	no_issues_found
TXN002	25,000.00 USD	flagged	fraud_risk_flagged (high_value, wire_transfer · 0.9)
TXN003	9,999.99 USD	approved	no_issues_found
TXN004	500.00 EUR	flagged	cross_border (DE)
TXN005	75,000.00 USD	flagged	fraud_risk_flagged (high_value, wire_transfer · 0.9)
TXN006	200.00 XYZ	rejected	invalid_currency
TXN007	-100.00 GBP	rejected	invalid_amount
TXN008	3,200.00 USD	approved	no_issues_found